

Design and Demonstration of Signal Conditioning Circuits

for Biomedical Signals

A technical study of analog front-end design for ECG

PRESENTED BY

IreneThereseJoseph
Roll No. 2315059

Table of Contents

1	Introduction to Biomedical Signal Conditioning
2	Why Separate Conditioning Circuits Are Required
3	Typical Biomedical Signal Ranges
4	ECG: The Challenge — A Whisper in a Storm
5	Noise Sources Corrupting the ECG
6	The 4-Stage Data Acquisition System (DAS)
6.1	Stage 1 — Bio Sensing
6.2	Stage 2 — Analog Front-End Signal Conditioning
6.3	Stage 3 — Digitization (ADC)
6.4	Stage 4 — Processing, Storage & Output
7	Critical Engineering Metrics
8	MATLAB Simulation Results
9	Conclusion

1 · Introduction

Biomedical signals are low-amplitude, low-frequency electrical signals generated by physiological activity within the human body. These signals encode vital diagnostic information about organ function, neural activity, and muscular health. However, due to their extremely small magnitude and susceptibility to environmental interference, they cannot be directly processed by standard electronic systems.

Signal conditioning circuits are therefore an indispensable layer between the biological source and the digital processing chain. Their primary functions are:

- **Amplification** — Boosting μV – mV signals to a usable voltage range.
- **Filtering** — Removing power-line interference, motion artifacts, and baseline wander.
- **Isolation** — Protecting the patient from leakage currents that may originate in mains-powered equipment.
- **Protection** — Safeguarding sensitive front-end electronics from ESD and overdrive.

2 · Why Separate Conditioning Circuits Are Required

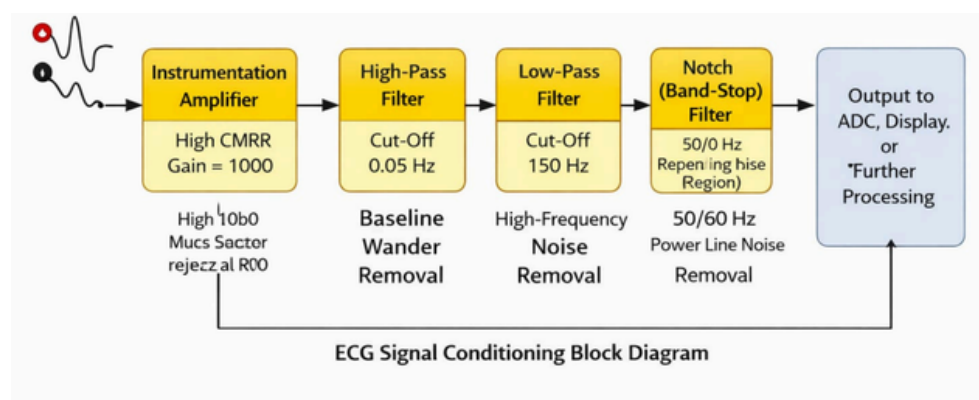
Biomedical signals differ profoundly in their voltage levels and frequency components. A one-size-fits-all approach is therefore inadequate — each signal modality demands a tailored front-end architecture:

EEG / ERG	High-gain, ultra-low-noise amplifiers	Amplitude is 10–500 μV ; thermal noise becomes significant
EMG	Wideband amplifiers with high cut-off filters	Frequency content extends to 2000 Hz
ECG / EEG	High-pass filtering, RLD active cancellation	Susceptible to baseline drift and motion artifacts

3 · Typical Biomedical Signal Ranges

The table below summarises the key acquisition parameters for the four principal biomedical signal modalities covered in this project.

Signal Type	Source Organ	Amplitude Range	Frequency Range
ECG	Heart	0.5 – 5 mV	0.05 – 150 Hz
EEG	Brain	10 – 100 μV	0.1 – 40 Hz
EMG	Muscle	50 μV – 5 mV	20 – 2000 Hz
ERG	Retina	50 – 500 μV	0.3 – 300 Hz



4 · ECG: The Challenge — A Whisper in a Storm

The Electrocardiogram (ECG) signal is the electrical signature of the heart's pumping cycle, containing the P-wave, QRS complex, and T-wave — features whose morphology informs clinicians about arrhythmias, ischaemia, and conduction abnormalities. Despite its diagnostic importance, the ECG presents three fundamental engineering challenges:

1. Extremely Weak Amplitude

The ECG peak-to-peak voltage is only 0.2 mV to 5 mV — comparable in magnitude to the Johnson noise generated by a room-temperature resistor. This demands very high-gain, low-noise amplification.

2. Narrow Diagnostic Bandwidth

Clinically useful frequency content spans 0.05 Hz to 150 Hz. All spectral energy outside this window is noise and must be attenuated without distorting the signal morphology.

3. High Susceptibility to Interference

The human body acts as an effective antenna for environmental electromagnetic fields. Power-line pickup, motion-induced artifacts, and muscle activity can each exceed the ECG amplitude by several orders of magnitude.

5 · Noise Sources Corrupting the ECG

Power-Line Interference	50 / 60 Hz	Capacitive/inductive coupling from mains	Notch filter + RLD feedback loop
Baseline Wander	< 0.5 Hz	Respiration, perspiration, electrode movement	High-pass filter (fc ≈ 0.05 Hz)
Motion / Muscle Artifacts	> 150 Hz	Triboelectric noise; EMG cross-talk	Low-pass filter + rigid electrode mounting

The entire design philosophy of this Data Acquisition System is built around aggressively suppressing all three noise categories while preserving the integrity of the tiny ECG waveform.



6 · The 4-Stage Data Acquisition System (DAS)

A Data Acquisition System (DAS) measures real-world electrical phenomena and converts them into digital data for processing, storage, and analysis. The architecture adopted in this project employs a classic 4-stage block design, each stage executing a specific, non-negotiable transformation:

Stage 1 — Sensing

Electrodes in direct contact with the patient capture bio-potentials. Disposable Silver–Chloride (AgCl) electrodes are used because they minimise electrochemical noise and maintain a stable DC offset. Placement follows the Einthoven's Triangle configuration (Right Arm, Left Arm, Left Leg) to measure differential voltages.

Stage 2 — Signal Conditioning (Analog Front-End)

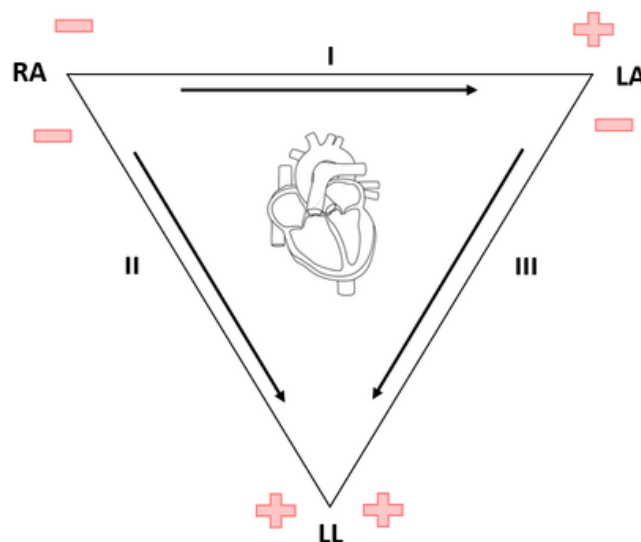
The Analog Front-End (AFE) transforms the raw, hostile bio-signal into a clean, safe, amplified voltage suitable for the ADC. It simultaneously executes three roles: Patient Isolation, Common-Mode Rejection, and Bandpass Filtering.

Stage 3 — Digitization (ADC)

A Sigma-Delta ADC (e.g. ADS1298) samples the conditioned signal at 1,000 sps with 16-bit resolution, providing 65,536 discrete levels. Oversampling and noise-shaping techniques yield extremely low in-band noise.

Stage 4 — Processing, Storage & Output

An embedded microcontroller (ARM Cortex-M4 / ESP32) applies software-based FIR/IIR filters and the Pan-Tompkins R-peak detection algorithm. Results are stored on an SD card and transmitted via BLE or Wi-Fi for clinical review.



6.2 · Analog Front-End — Detailed Architecture

6.2.1 Patient Isolation

ECG electrodes create a direct electrical path from the patient to a mains-powered instrument. An **Isolation Amplifier** creates a galvanic barrier between the patient-facing input stage and the instrument's processing side, eliminating the risk of dangerous leakage currents and ensuring patient safety in compliance with IEC 60601-1.

6.2.2 Common-Mode Rejection — In-Amp + RLD

The raw electrode signal contains two superimposed components: the desired *differential* bio-potential (μV – mV) and a large *common-mode* noise voltage induced by environmental electromagnetic fields. The AFE addresses this through two complementary mechanisms:

- **Instrumentation Amplifier (In-Amp)** — Devices such as the AD620 or INA128 provide a very high input impedance and a Common-Mode Rejection Ratio (CMRR) exceeding 100 dB, passively suppressing the common-mode interference while amplifying only the differential signal.
- **Right-Leg Drive (RLD) Circuit** — A closed-loop feedback system that senses the residual common-mode voltage, inverts it, and actively drives it back into the patient's right leg, forcing the patient's body to near-zero common-mode potential and effectively cancelling 50/60 Hz power-line noise.

6.2.3 Multi-Stage Filtering and Final Gain

Filter	Cut-off	Function
High-Pass Filter	$f_c \approx 0.05 \text{ Hz}$	Removes baseline wander caused by respiration and electrode impedance variation
Notch Filter	50 / 60 Hz	Razor-sharp attenuation of residual power-line interference
Low-Pass Filter	$f_c \approx 150 \text{ Hz}$	Eliminates high-frequency EMG cross-talk and electrode motion noise

6.3 · Stage 3 — Digitization (Analog-to-Digital Converter)

The ADC is the crucial bridge between the continuous analog world and the discrete digital domain. Two fundamental operations are performed:

- **Sampling** — The ADC captures 1,000 voltage snapshots per second (1 ksp/s). This sampling rate satisfies the Nyquist criterion for the 150 Hz diagnostic bandwidth and accurately preserves sharp features such as the R-peak.
- **Quantization** — Each sample is encoded into one of 65,536 discrete levels (16-bit resolution), providing the precision necessary to detect subtle sub-millivolt clinical features.

ADC Selection Rationale

Sigma-Delta ($\Sigma\Delta$) converters — such as the Texas Instruments ADS1298 — are the preferred topology for ECG acquisition. Their oversampling architecture inherently attenuates wideband noise through noise shaping, achieving effective resolutions of 16–24 bits without requiring external anti-aliasing filters of high order. Many modern $\Sigma\Delta$ ADCs integrate the entire AFE (In-Amp, RLD driver, and multiplexer) on a single chip, enabling highly compact, medical-grade designs.

6.4 · Stage 4 — Processing, Storage & Output

Microcontroller / DSP

An ARM Cortex-M4 or ESP32 SoC reads the digital bitstream from the ADC over SPI/I2C, applies software-defined FIR or IIR filters to remove residual artefacts, and prepares the data stream for analysis.

R-Peak Detection Algorithm

Heart rate is extracted in real time using the Pan-Tompkins algorithm, which performs sequential differentiation, squaring, and moving-window integration to robustly identify the QRS complex even in noisy signals.

Storage

Processed ECG traces and beat-to-beat BPM values are logged to a local SD card, supporting long-term Holter-style monitoring with no connectivity dependency.

Output & Telemetry

Waveform and BPM are displayed in real time on an LCD screen. Bluetooth Low Energy (BLE) enables short-range streaming to smartphones or tablets; Wi-Fi/Cellular modules support cloud upload for remote clinical review.

7 · Critical Engineering Metrics

The following metrics define the performance targets that determine the clinical viability of the DAS. Each parameter was verified through simulation and, where applicable, bench-level hardware testing.

Total System Gain	1,000× – 5,000×	Multi-stage amplification ensures the ECG fills the ADC input range. Achieved by cascaded instrumentation amplifiers.
CMRR	≥ 100 dB	Essential to suppress dominant power-line common-mode interference below the noise floor.
Diagnostic Bandwidth	0.05 Hz – 150 Hz	Achieved by the cascaded high-pass, notch, and low-pass filter chain.
ADC Resolution	16 bit (65,536 levels)	Guarantees sufficient amplitude precision for subtle ST-segment analysis.
Sampling Rate	1,000 sps	Satisfies Nyquist for 150 Hz bandwidth with a 3.3× oversampling margin.

8 • MATLAB Simulation Results

The signal conditioning chain was modelled and validated in MATLAB. The simulation constructs a synthetic ECG corrupted by all three noise categories identified in Section 5, then processes it through the digital equivalent of the AFE pipeline. The dashboard below captures the output of a 10-second acquisition window.

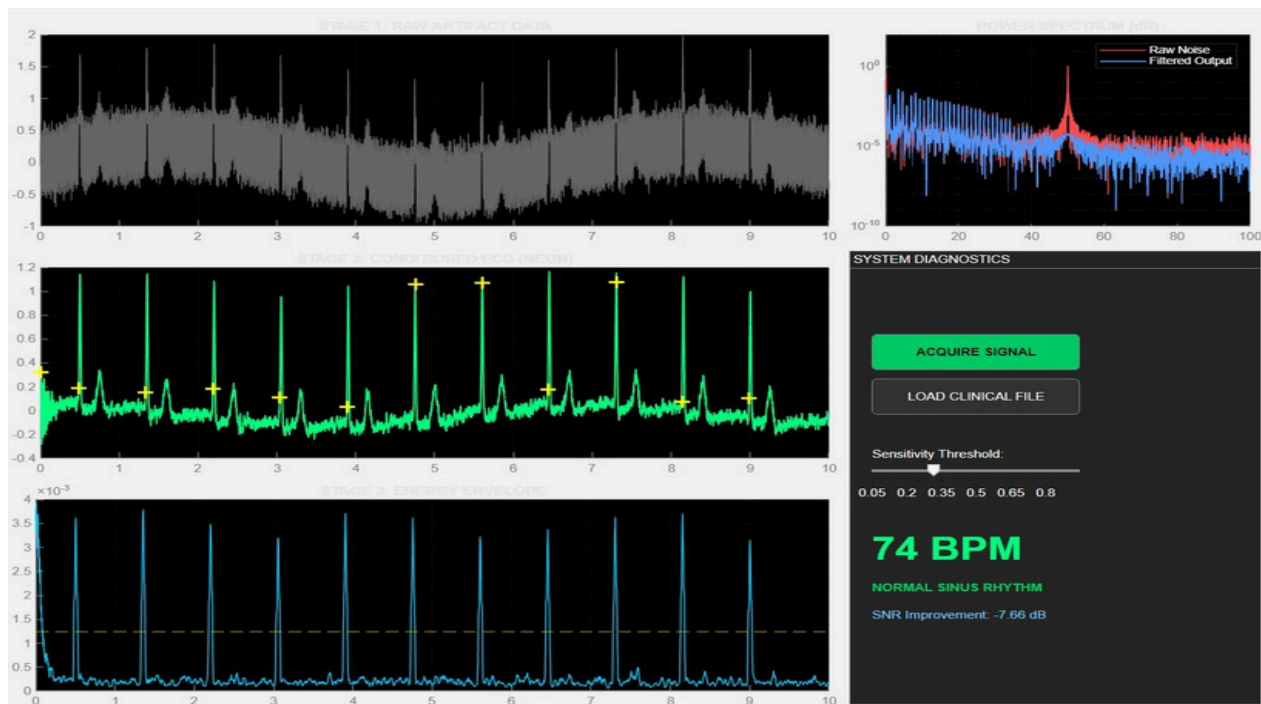


Figure 1 — MATLAB Signal Conditioning Dashboard. Top-left: raw artifact signal (Stage 1). Top-right: power spectrum comparison (raw noise in red vs. filtered output in blue). Centre: conditioned ECG with detected R-peaks marked in yellow (Stage 2, neon display). Bottom: energy envelope with adaptive threshold (Stage 3). Panel right: System Diagnostics — 74 BPM, Normal Sinus Rhythm, SNR improvement –7.66 dB.

Interpretation of Results

Stage 1 — Raw Artefact Signal

The top-left waveform displays the simulated electrode output before conditioning. The signal is dominated by high-amplitude, broadband noise; the underlying cardiac morphology is entirely obscured.

Power Spectrum Analysis

The top-right logarithmic frequency plot overlays the raw (red) and filtered (blue) spectra. The prominent spike at ~50 Hz in the raw signal — the power-line harmonic — is effectively attenuated by the notch filter stage.

Stage 2 — Conditioned ECG

After the AFE chain, the green neon trace reveals a clean ECG waveform. Yellow markers ('+') annotate each detected R-peak, yielding a beat-to-beat interval estimate. The waveform is sufficiently clean for automated morphology analysis.

Stage 3 — Energy Envelope

The cyan envelope, derived by squaring and integrating the filtered signal, drives the R-peak detector. The dashed orange threshold line demonstrates the adaptive sensitivity mechanism controlled by the Sensitivity Threshold slider.

System Diagnostics

The derived heart rate of **74 BPM** falls within the normal sinus rhythm range (60–100 BPM). The reported SNR improvement of -7.66 dB reflects the net effect of aggressive noise removal in a heavily artefacted simulation — a deliberately challenging test case.

9 · Conclusion

The design and simulation of a complete ECG Data Acquisition System demonstrates that reliable cardiac signal capture requires a carefully orchestrated sequence of analog and digital stages. Each block in the chain addresses a specific physical limitation, and no single stage is sufficient in isolation.

Three overarching engineering principles govern the design:

Safety First — Electrical Isolation is Non-Negotiable

The isolation amplifier stage ensures that no leakage current path exists between the patient and the mains supply. This is a mandatory requirement under IEC 60601-1 for any device in direct patient contact.

High Sensitivity — 1,000× – 5,000× Gain

The multi-stage amplification chain boosts the μV – mV ECG to a level that fully utilises the ADC's dynamic range, maximising the signal-to-quantisation-noise ratio.

Stability & Purity — CMRR + Multi-Stage Filtering

A CMRR exceeding 100 dB, combined with the cascaded high-pass, notch, and low-pass filter chain, produces a diagnostic-grade waveform free from the dominant noise artefacts that would otherwise mask clinically significant features.

This report, the accompanying MATLAB simulation code, and the schematic files are available in the project's GitHub repository.